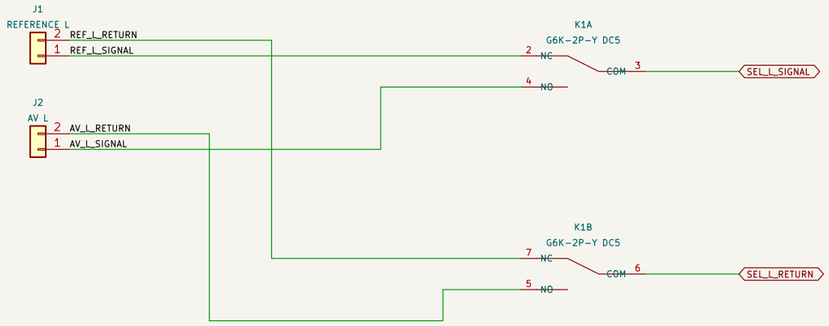
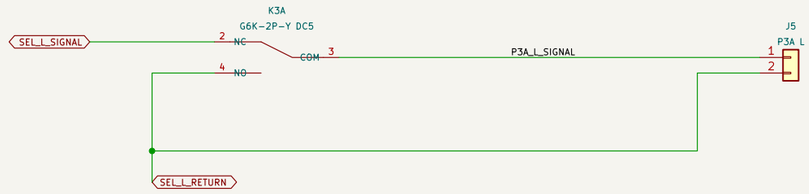
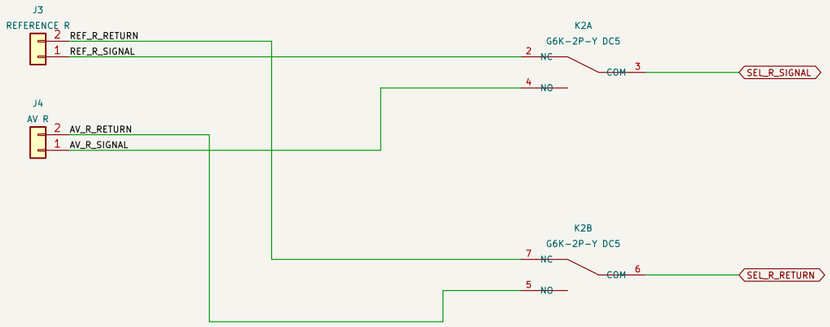
CONTACTS ONLY IN THE THROUGH SIGNAL PATH

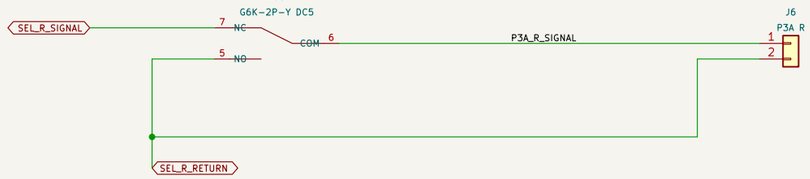
K1/K2 OFF = REFERENCE. ON = AV. K3 OFF = PASS. ON = output shorted to its own return.
All RCA shells must be insulated from the enclosure. Keep each signal with its corresponding return.





L channel: no common GND





R channel: no common GND

Sensing taps are on SEL_* before K3. No permanent connection exists to inactive inputs.
The P3A may join its own L/R returns externally: this board does not join them.
Unpowered pass-through is intentional: abrupt controller loss is not a guaranteed mute.

12V input and 5V buck supply

File: Power.kicad_sch

ATtiny412 and relay drivers

File: Control.kicad_sch

Isolated left detector

File: Detector_L.kicad_sch

Isolated right detector

File: Detector_R.kicad_sch

ULN2003A/G6K pickup margin requires bench validation.
Low-voltage controller only. No mains on this PCB.

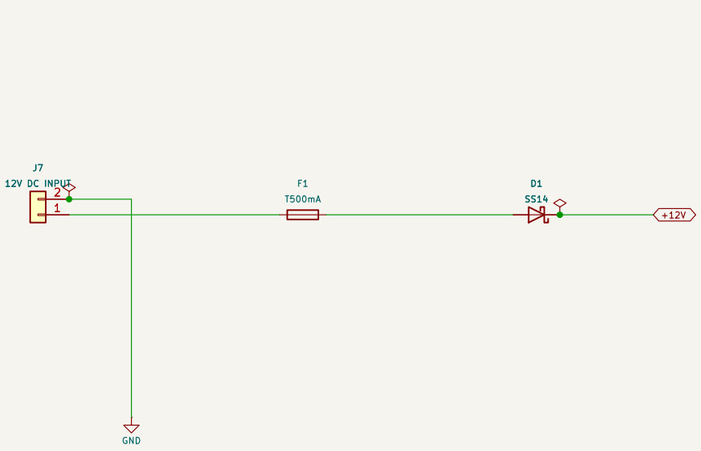
Sheet: /
File: InputSwitch.kicad_sch

Title: Audio source selector and output mute

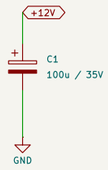
Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/5

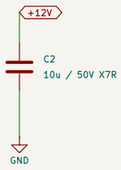
ISOLATED REGULATED 12V WALL ADAPTER: 12V +/−2%, >=0.5A

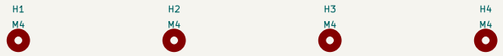
DC input only. G7L mains contacts and all 230V wiring are chassis−mounted off−board.
PE remains continuous and is never switched. Supply GND is the controller ground only.

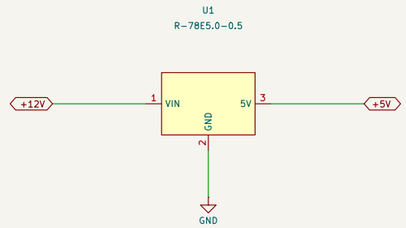


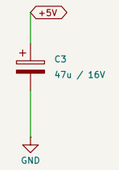


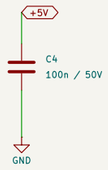












D1 reverse−polarity blocking; F1 limits fault current; D2 clamps short input transients.
Not a sustained−overvoltage disconnect. Use only the specified regulated adapter.
SS14 drop reduces the coil/module feed slightly below the adapter voltage.

5V supply: 500mA capacity; 3 relay coils total about 63mA nominal.
Allow 100mA total 5V load, plus two isolated detector modules on 12V.
R−78E total output capacitance must stay below 220uF.
Input TVS/fuse coordination and ripple/EMI need bench/layout validation.







NO PCB MAINS. Do not update the old PCB from this schematic until layout work is authorised.

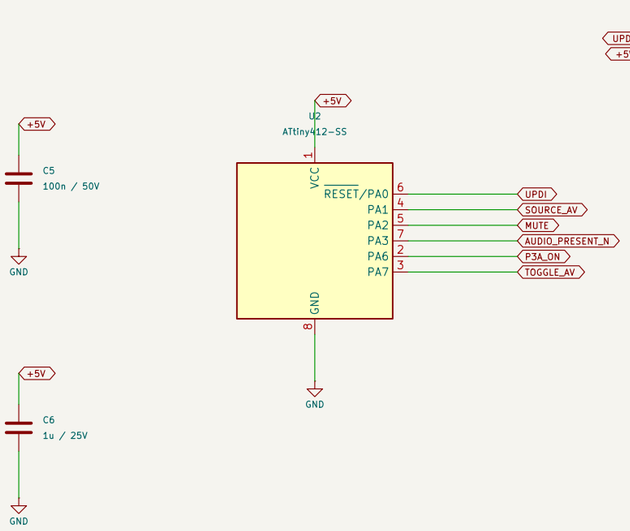
ULN2003A/G6K pickup margin requires bench validation.
Low−voltage controller only. No mains on this PCB.

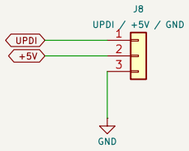
Sheet: /12V input and 5V buck supply/
File: Power.kicad_sch

Title: 12V input and 5V buck supply

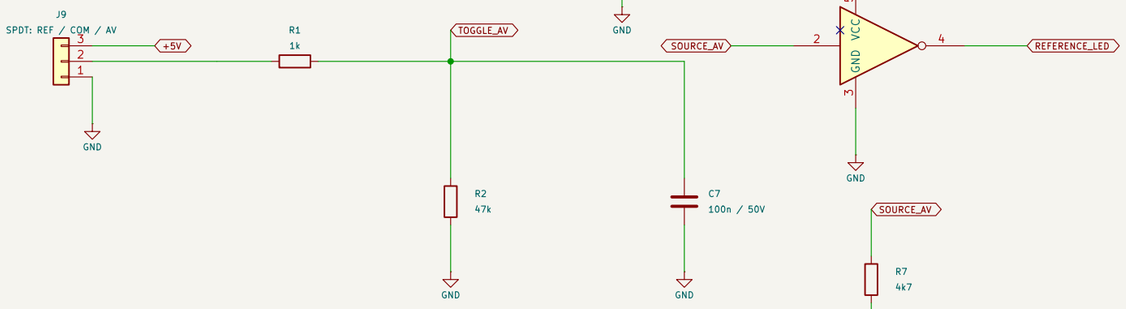
Size: A3	Date: 2026−10−01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 2/5

FIVE GPIOs; PA0 REMAINS UPDI

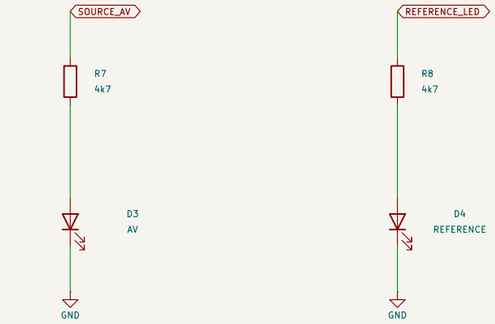




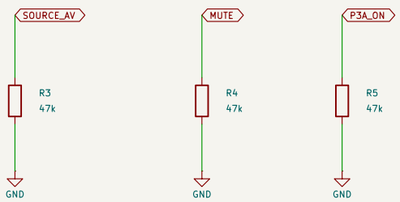
J8 pin 1 UPDI; pin 2 target-voltage reference; pin 3 GND.
Programmer must not back-power the board. BOD 4.2V + watchdog in firmware.

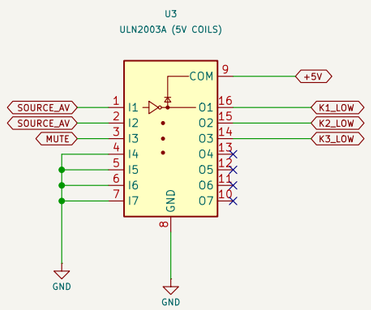


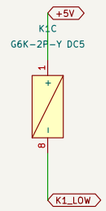
Open/disconnected switch -> REFERENCE.
100us contact-driven RC; 4.8ms with open switch.
Firmware debounce >=20ms.

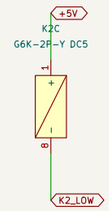


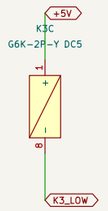
LEDs indicate source COMMAND.
They do not verify relay contacts.

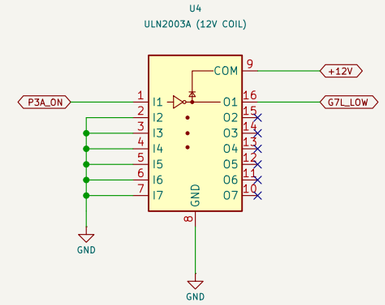






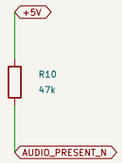


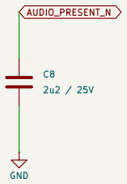




Driver inputs tied low when MCU is reset.
Unused ULN inputs grounded; outputs genuinely unused.
No parallel output channels.

G7L-2A-T DC12: J10 pin 1 coil+; pin 2 low-side.







Optocoupler wired-OR: LOW = recent audio. 103ms nominal release RC.
Firmware must require 500ms persistence, then time 10min silence.
Blank detection at startup, source changes and after power-off.
Do not count a single edge/click as continuous audio.

ULN2003A/G6K pickup margin requires bench validation.
Low-voltage controller only. No mains on this PCB.

Sheet: /ATtiny412 and relay drivers/
File: Control.kicad_sch

Title: ATtiny412 and relay drivers

Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 3/5

L CHANNEL — INDEPENDENT GALVANIC ISOLATION

Isolated return is tied ONLY to SEL_L_RETURN, after its source-selection relay.
No detector wire, resistor, capacitor or shield connects this return to control GND or the other channel.

U10
R12P12S/P
+12V
VIN+
VIN-
VOUT+
VOUT-
ISO_L_12V
SEL_L_RETURN
GND

C20
10u / 50V X7R

R20
1k2 / 0.5W

R21
100R

C21
47u / 25V

C22
100n / 50V

U5C
LM358N

R22
1k2

D20
BZX55C5V1

C23
33u / 16V

1k2 preload holds the isolated converter above 10% load.
100R + 47u filters its switching ripple. No capacitor crosses isolation.
Module input range 10.8–13.2V: verify at the protected +12V rail.

C24
100n FILM / 50V

R23
100k

R24
470k

U5A
LM358N

R25
100k

R26
1k

C25
33u / 16V

R27
10k

R29
1M

U5B
LM358N

R28
100k

DETECT_PULSE

R30
10k

R31
100k

R32
1k8

U6
VO617A-3

R33
1k

Q2
BC549

SEL_L_SIGNAL
SENSE_INPUT
BIAS_L
AUDIO_PRESENT_N

CONTROL SIDE

Optocoupler collector is the only
logic interface. Both channels
wire-OR into one active-low input.

D21
1N4148

D22
1N4148

Branch clamps only;
never in the main audio path.

Nominal midband input impedance ~570k. AC gain ~101 x 470/570 = 83.3.
Nominal trigger ~4mVrms at 1kHz; tuning/bench value, not a guaranteed noise threshold.
At 2Vrms the detector saturates intentionally; it is outside the through audio path.
100k limits loading when unpowered. Only ordinary SELV signals belong here.

ULN2003A/G6K pickup margin requires bench validation.
Low-voltage controller only. No mains on this PCB.

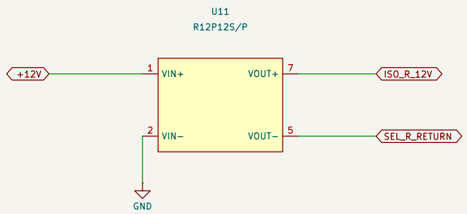
Sheet: /Isolated left detector/
File: Detector_L.kicad_sch

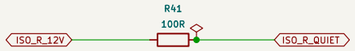
Title: Isolated left detector

Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 4/5

R CHANNEL — INDEPENDENT GALVANIC ISOLATION

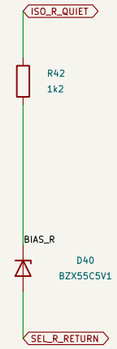
Isolated return is tied ONLY to SEL_R_RETURN, after its source-selection relay.
No detector wire, resistor, capacitor or shield connects this return to control GND or the other channel.





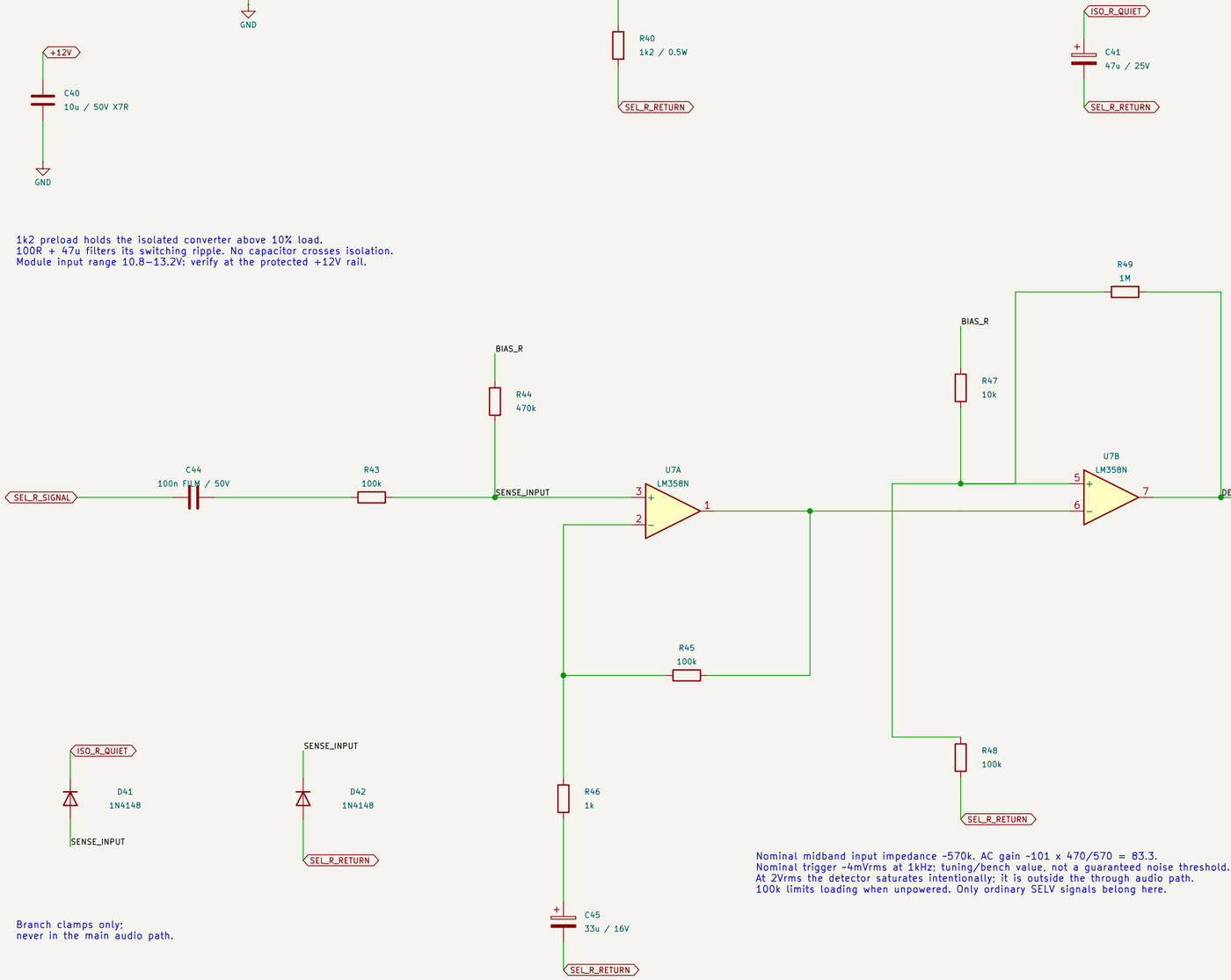








1k2 preload holds the isolated converter above 10% load.
100R + 47u filters its switching ripple. No capacitor crosses isolation.
Module input range 10.8–13.2V: verify at the protected +12V rail.



CONTROL SIDE

Optocoupler collector is the only
logic interface. Both channels
wire-OR into one active-low input.

Branch clamps only;
never in the main audio path.

Nominal midband input impedance ~570k. AC gain ~101 x 470/570 = 83.3.
Nominal trigger ~4mVrms at 1kHz; tuning/bench value, not a guaranteed noise threshold.
At 2Vrms the detector saturates intentionally; it is outside the through audio path.
100k limits loading when unpowered. Only ordinary SELV signals belong here.

ULN2003A/G6K pickup margin requires bench validation.
Low-voltage controller only. No mains on this PCB.

Sheet: /Isolated right detector/
File: Detector_R.kicad_sch

Title: Isolated right detector

Size: A3	Date: 2026-10-01	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 5/5